

The Torque Hinge

One rotational balance, three physical origins of torque

Working rule: Begin with the chosen axis and $\sum \tau_{\text{ext}} = dL/dt$. For a rigid body rotating about a fixed axis with constant moment of inertia, this becomes $\sum \tau = I_{\text{rot}}\alpha$. The equation connects torque to angular acceleration; it does not say what interaction supplies the torque.

Name: _____

Class: _____

Companion task to *The Placeholder Force*

1. Gravity: a rod released from horizontal

A uniform rod of length 1.20 m and mass 3.0 kg is hinged at one end and released from rest while horizontal. Take $g = 9.8 \text{ m s}^{-2}$ and $I_{\text{rot}} = \frac{1}{3}ML^2$ about the hinge.

A. Represent force, axis and lever arm

Draw the rod, hinge, centre of mass and weight. Mark the rotation axis. Draw the perpendicular distance from the axis to the line of action of the weight. Include the hinge force, even though it will not appear in the torque equation about the hinge.

B. Supply the gravitational torque

At the instant of release, calculate

$$\tau_g = Mg \frac{L}{2}.$$

State its rotational direction. Explain why this torque comes from the gravitational interaction rather than from $\tau = I_{\text{rot}}\alpha$.

C. Apply the rotational hinge

Use $\sum \tau = I_{\text{rot}}\alpha$ to find the initial angular acceleration. Show that the rod’s mass cancels. Does that cancellation mean gravity and rotational inertia are the same physical property?

D. Interrogate the geometry

Why does the hinge force produce zero torque about the hinge? How would the gravitational torque change as the rod approaches the downward vertical? Which part of the model contains this angular dependence?

2. Friction: a brake slows a flywheel

A solid disc of mass 4.0 kg and radius 0.25 m rotates at 18 rad s^{-1} . A brake pad presses normally on its rim with $N = 10 \text{ N}$. The kinetic-friction coefficient is $\mu_k = 0.30$. Use $I_{\text{rot}} = \frac{1}{2}MR^2$ and neglect axle friction.

A. Represent the contact interaction

Sketch the disc, axis, pad, normal force and friction force. Draw the friction force in the direction that opposes the motion of the rim. Mark the radius that converts the friction force into torque.

B. Close the model with friction

Use the contact law $f_k = \mu_k N$ and the geometry $\tau_f = f_k R$ to calculate the braking torque. Which of these relations is empirical, and which is geometrical?

C. Join torque to angular motion

Calculate I_{rot} , the angular acceleration and the stopping time. Verify the result using angular impulse,

$$\tau \Delta t = \Delta L = I_{\text{rot}} \Delta \omega.$$

D. Interrogate loss and system boundary

Where does the disc’s rotational kinetic energy go? Why is the disc’s angular momentum not conserved? Name a larger system for which total angular momentum could be conserved.

3. Magnetic interaction: torque on a coil

A 50-turn rectangular coil of area $8.0 \times 10^{-3} \text{ m}^2$ carries current 0.60 A in a uniform field of 0.25 T. Its rotational inertia is $I_{\text{rot}} = 2.0 \times 10^{-2} \text{ kg m}^2$. Initially the coil’s magnetic moment is perpendicular to the field.

A. Represent the magnetic couple

Draw the coil, field, current direction and normal vector to the coil. Show the opposite magnetic forces on two sides. Explain how their net force can be zero while their net torque is non-zero.

B. Supply the magnetic torque

Use

$$\tau_B = NIAB \sin \theta$$

to calculate the initial torque, where I here denotes electric current. Identify the coupling parameters and the geometrical factor. What happens at $\theta = 0$?

C. Apply the rotational hinge

Find the initial angular acceleration. Predict the effect of reversing the current, doubling the field, and doubling I_{rot} . Distinguish carefully between electric current I and rotational inertia I_{rot} .

D. Interrogate the motor model

Why is the calculated angular acceleration only instantaneous? What additional effects would matter once the coil begins to rotate in a practical motor?

Synthesis: the rotational balance is shared; the torque law supplies the physics

1. Complete the comparison.

Case	Chosen axis	Physical torque law	Coupling or parameter	What I_{rot} represents
Gravity				
Friction				
Magnetic				

2. Develop the central claim.

Respond to:

“The symbol τ is a rotational placeholder: it records the moment of an interaction without identifying the interaction.”

Use all three cases. Include the terms *axis*, *lever arm*, *moment of inertia*, *external torque*, *angular momentum*, *couple*, and *empirical closure*. Explain why $\tau = I_{\text{rot}}\alpha$ is less general than $\sum \tau_{\text{ext}} = dL/dt$.

Teacher guide and indicative answers

The Torque Hinge

1. Gravity: torque from a field interaction

Representation. Weight Mg acts at the centre of mass, $L/2$ from the hinge. The hinge reaction passes through the chosen axis and therefore has zero moment about that axis, although it remains a real force.

Torque.

$$\tau_g = Mg \frac{L}{2} = (3.0)(9.8)(0.60) = 17.64 \text{ N m}.$$

Its direction is the direction in which the rod begins to fall.

Rotational inertia.

$$I_{\text{rot}} = \frac{1}{3}ML^2 = \frac{1}{3}(3.0)(1.20)^2 = 1.44 \text{ kg m}^2.$$

Hence

$$\alpha = \frac{\tau_g}{I_{\text{rot}}} = 12.25 \text{ rad s}^{-2}.$$

Algebraically,

$$\alpha = \frac{MgL/2}{ML^2/3} = \frac{3g}{2L}.$$

The cancellation occurs because both gravitational torque and rotational inertia scale with the rod's mass. It does not identify gravitational mass with moment of inertia; they are different quantities with different dimensions and roles.

Angular dependence. If θ is the angle between the rod and the downward vertical, $\tau_g = Mg(L/2) \sin \theta$. It is maximal when horizontal and zero when vertical. This dependence belongs to the torque law and geometry, not to $\tau = I_{\text{rot}}\alpha$.

Diagnostic question. Changing the axis changes both the lever arm and I_{rot} . Students should not treat torque or moment of inertia as properties of the body alone without an axis.

2. Friction: torque from a contact law

Friction and torque.

$$f_k = \mu_k N = (0.30)(10) = 3.0 \text{ N}, \quad \tau_f = f_k R = 0.75 \text{ N m}.$$

The minus sign is introduced by the chosen rotational direction: the torque opposes ω .

The relation $f_k = \mu_k N$ is an empirical contact model. The relation $\tau = rF_{\perp}$ is geometrical.

Neither is supplied by rotational dynamics.

Moment of inertia and deceleration.

$$I_{\text{rot}} = \frac{1}{2}MR^2 = \frac{1}{2}(4.0)(0.25)^2 = 0.125 \text{ kg m}^2,$$

$$\alpha = -\frac{0.75}{0.125} = -6.0 \text{ rad s}^{-2}.$$

Thus

$$t_{\text{stop}} = \frac{18}{6.0} = 3.0 \text{ s}.$$

Angular impulse check.

$$L_i = I\omega = 0.125(18) = 2.25 \text{ kg m}^2 \text{ s}^{-1},$$

$$\tau \Delta t = (-0.75)(3.0) = -2.25 \text{ kg m}^2 \text{ s}^{-1}.$$

Energy and boundary. Initial rotational energy is $\frac{1}{2}I\omega^2 = 20.25 \text{ J}$, transferred mainly into thermal energy of the pad, disc and surroundings. The disc alone experiences an external torque, so its angular momentum is not conserved. A sufficiently enlarged system including brake, mounting and Earth can conserve total angular momentum.

3. Magnetic interaction: a force couple

Physical picture. Opposite sides of the current loop experience equal and opposite magnetic forces. Their vector sum can vanish while their separated lines of action form a couple and produce a torque.

Magnetic torque. At $\theta = 90^\circ$,

$$\tau_B = NIAB = (50)(0.60)(8.0 \times 10^{-3})(0.25) = 6.0 \times 10^{-2} \text{ N m}.$$

The turns N , current I , area A and field B determine the interaction strength; $\sin \theta$ supplies the orientation. At $\theta = 0$, the magnetic moment is aligned with the field and the torque is zero.

Angular acceleration.

$$\alpha = \frac{0.060}{2.0 \times 10^{-2}} = 3.0 \text{ rad s}^{-2}.$$

Reversing current reverses the torque. Doubling B doubles τ and α . Doubling I_{rot} halves α for the same torque.

Instantaneous limitation. As the coil turns, θ changes, so the torque changes. A practical motor also involves commutation, back emf, resistance, magnetic field geometry, bearing friction and load torque. Constant current and uniform field are closure assumptions for the simplified model.

Notation warning. In $NIAB$, I is current. In $I_{\text{rot}}\alpha$, the moment of inertia is a different physical quantity. The shared letter is another form of notational compression.

Teacher synthesis: what should survive the calculations?

Indicative comparison.

Case	Torque source	Closure	Rotational resistance
Gravity	$\mathbf{r} \times M\mathbf{g}$	field law and centre-of-mass geometry	I_{rot} about hinge
Friction	$\mathbf{r} \times \mathbf{f}$	$f_k = \mu_k N$	I_{rot} about axle
Magnetic	magnetic force couple	$\tau = NIAB \sin \theta$	coil and rotor I_{rot}

Core interpretation. The general dynamical statement is

$$\sum \boldsymbol{\tau}_{\text{ext}} = \frac{d\mathbf{L}}{dt}.$$

The familiar scalar form $\sum \tau = I_{\text{rot}}\alpha$ requires a rigid body, a fixed axis and constant I_{rot} . It is therefore a specialised bridge. In every panel, additional physics determines the torque: gravity and geometry, a friction law, or a magnetic interaction. The moment of inertia determines how much angular acceleration that supplied torque produces.

Expected conclusion. The same torque slot can contain different mechanisms. The equation of rotational dynamics transmits their consequence into angular motion; it neither identifies the interaction nor chooses the axis.